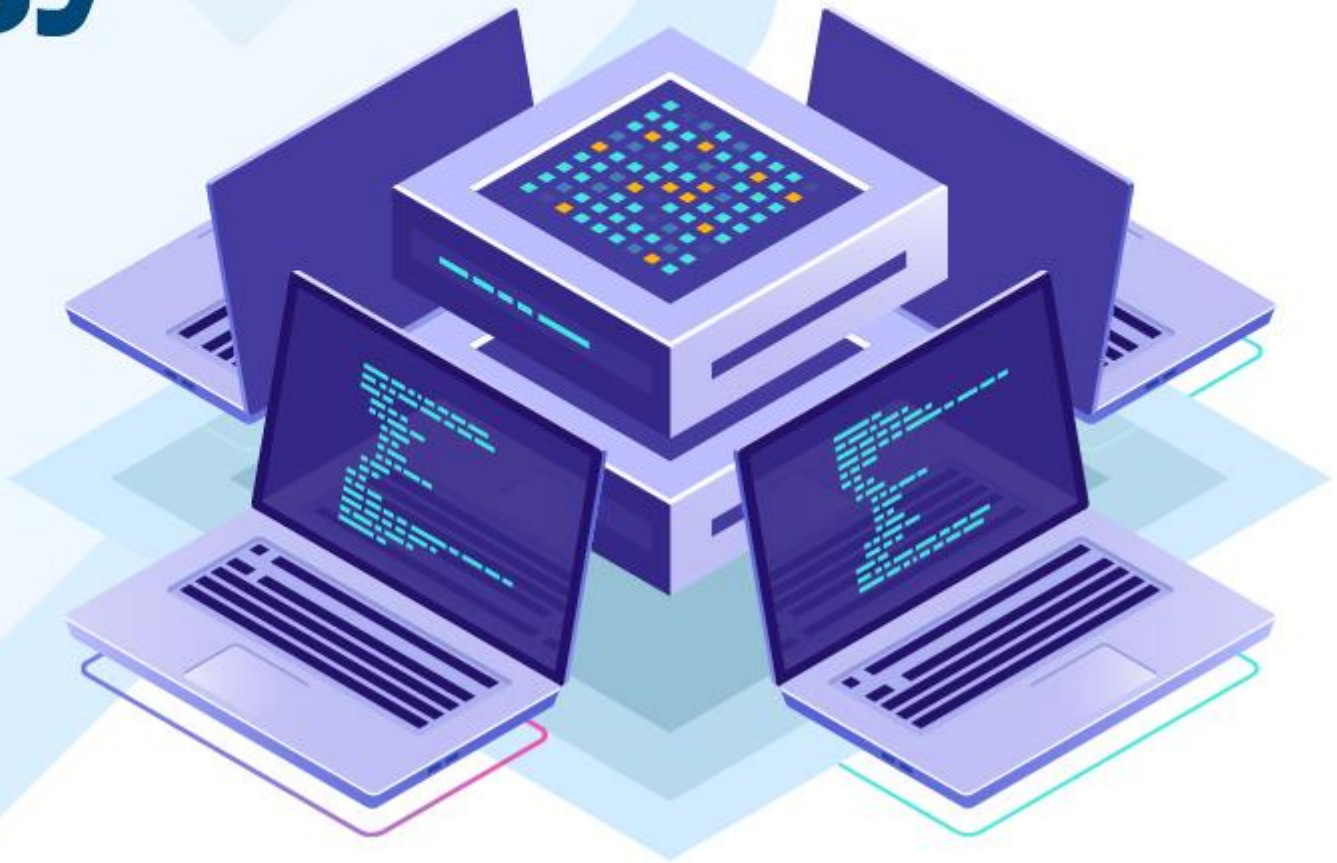
Information Technology

Fall Semester
2026 - 2027

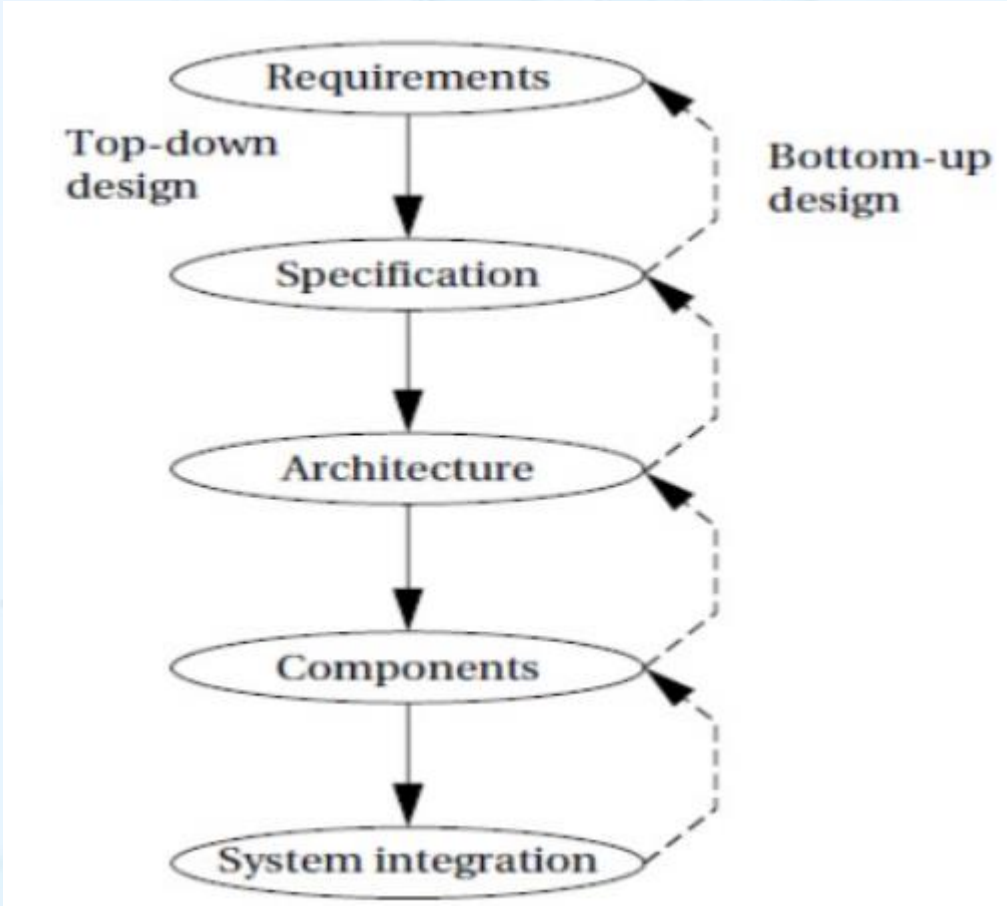
- Embedded Systems
- **Dr. Yasmine Mahmoud**



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Embedded System Design Process:

- The major steps in the embedded system design process.
- In this top–down view, we start with the system requirements.



Embedded System Design Process:

Requirements:

- Clearly, before we design a system, we must know what we are designing.
  - The initial stages of the design process capture this information for use in creating the architecture and components.
- We generally proceed in two phases:
  - First, we gather requirements, and we refine the requirements into a specification that contains enough information to begin designing the system architecture.

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Embedded System Design Process:

Requirements:

- Requirements may be functional or nonfunctional.
- We must capture the basic functions of the embedded system, but functional description is often not sufficient.
- Typical nonfunctional requirements (Embedded Systems Challenges) include:
  - Performance.
  - Cost.
  - Physical size and weight.
  - Power consumption.

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Non-Functional Requirements

1. Performance:

- The speed of the system is often a major consideration both for the usability of the system and for its ultimate cost.
- Performance may be a combination of:
  - soft performance metrics such as approximate time to perform a user-level function and,
  - hard deadlines by which a particular operation must be completed.

Non-Functional Requirements

2. Cost:

- The target cost or purchase price for the system is almost always a consideration.
- Cost typically has two major components:
  - manufacturing cost includes the cost of components and assembly;
  - nonrecurring engineering (NRE) costs include the personnel and other costs of designing the system.

Non-Functional Requirements

3. Physical size and weight:

- The physical aspects of the final system can vary greatly depending upon the application.
- An industrial control system for an assembly line may be designed to fit into a standard-size rack with no strict limitations on weight.
- A handheld device typically has tight requirements on both size and weight that can ripple through the entire system design.

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Non-Functional Requirements

4. Power consumption:

- Power, is important in battery-powered systems and is often important in other applications as well.
- Power can be specified in the requirements stage in terms of battery life.
- The customer is unlikely to be able to describe the allowable wattage.

A sample requirements form

- A sample requirements form that can be filled out at the start of the project.
- We can use the form as a checklist in considering the basic characteristics of the system.

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A sample requirements form

- Let's consider the entries in the form:
  - **Name:**
    - This is simple but helpful.
    - Giving a name to the project not only simplifies talking about it to other people but can also crystallize the purpose of the machine.
  - **Purpose:**
    - This should be a brief one or two lines description of what the system is supposed to do.
    - If you can't describe the essence of your system in one or two lines, chances are that you don't understand it well enough.

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A sample requirements form

- Let's consider the entries in the form:
  - **Inputs and outputs:**

    These two entries are more complex than they seem.

    – The inputs and outputs to the system encompass a wealth of detail:
      - Types of data: Analog electronic signals? Digital data? Mechanical inputs?
      - Data characteristics: Periodically arriving data, such as digital audio samples? Occasional user inputs? How many bits per data element?
      - Types of I/O devices: Buttons? Analog/digital converters? Video displays?

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A sample requirements form

- Let's consider the entries in the form:
  - **Functions:**
  - This is a more detailed description of what the system does.
  - A good way to approach this is to work from the inputs to the outputs:
    - When the system receives an input, what does it do?
    - How do user interface inputs affect these functions?
    - How do different functions interact?

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A sample requirements form

- Let's consider the entries in the form:
  - **Performance:**
    - Many embedded computing systems spend at least some time controlling physical devices or processing data coming from the physical world.
    - In most of these cases, the computations must be performed within a certain time frame.
    - It is essential that the performance requirements be identified early since they must be carefully measured during implementation to ensure that the system works properly.

A sample requirements form

- Let's consider the entries in the form:
  - **Manufacturing cost:**
    - This includes primarily the cost of the hardware components. Even if you don't know exactly how much you can afford to spend on system components, you should have some idea of the eventual cost range.
    - Cost has a substantial influence on architecture:
      - A machine that is meant to sell at $10 most likely has a very different internal structure than a $100 system.

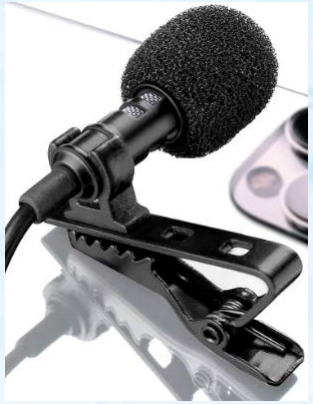
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A sample requirements form

- Let's consider the entries in the form:
  - **Power:**
    - Similarly, you may have only a rough idea of how much power the system can consume, but a little information can go a long way.
    - Typically, the most important decision is whether the machine will be battery powered or plugged into the wall.
    - Battery-powered machines must be much more careful about how they spend energy.

A sample requirements form

- Let's consider the entries in the form:
  - **Physical size and weight:**
    - You should give some indication of the physical size of the system to help guide certain architectural decisions.
    - A desktop machine has much more flexibility in the components used than, for example, a lapel mounted voice recorder.



A sample requirements form



GPS MODULE:

REQUIREMENTS FORM OF GPS MOVING MAP MODULE:

- Name : GPS moving map
- Purpose: Consumer-grade moving map for driving use
- Inputs : Power button, two control buttons
- Outputs : Back-lit LCD display 400 _ 600
- Functions : Uses 5-receiver GPS system;
  three user-selectable resolutions;
  always displays current latitude and longitude
- Performance: Updates screen within 0.25 seconds upon movement
- Manufacturing cost: $30
- Power: 100mW
- Physical size and weight: No more than 2” _ 6, ” 12 ounces

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THANK YOU FOR WATCHING

QUESTIONS?

